

Accepted Article

Title: Mitigating the Shielding Effect of Ether Oxygen in Poly(ethylene glycol) (PEG) on Boron (B) Atoms in B Doped PEG (B-PEG) Hybrid Polymer Electrolyte via Introducing Siloxane Spacers

Authors: Ran Zhang, Zhaoyang Wei, Weiwei Lei, Tao Jiang, Qunchao Zhang, and Dean Shi

This manuscript has been accepted after peer review and appears as an Accepted Article online prior to editing, proofing, and formal publication of the final Version of Record (VoR). This work is currently citable by using the Digital Object Identifier (DOI) given below. The VoR will be published online in Early View as soon as possible and may be different to this Accepted Article as a result of editing. Readers should obtain the VoR from the journal website shown below when it is published to ensure accuracy of information. The authors are responsible for the content of this Accepted Article.

To be cited as: *ChemElectroChem* 10.1002/celc.202000784

Link to VoR: <https://doi.org/10.1002/celc.202000784>

Mitigating the Shielding Effect of Ether Oxygen in Poly(ethylene glycol) (PEG) on Boron (B) Atoms in B Doped PEG (B-PEG) Hybrid Polymer Electrolyte via Introducing Siloxane Spacers

Ran Zhang, Zhaoyang Wei*, Weiwei Lei, Tao Jiang, Qunchao Zhang and Dean Shi*

Hubei Collaborative Innovation Center for Advanced Organic Chemical Materials, Ministry of Education Key Laboratory for the Green Preparation and Application of Functional Materials, Hubei Key Laboratory of Polymer Materials, Faculty of Materials Science and Engineering, Hubei University, Wuhan 430062, China.

Abstract: Doping Boron (B) atoms into Poly(ethylene glycol) (PEG) based solid polymer electrolytes (B-PEG) is believed an attractive method to increase the lithium ion transference number (t_{Li}^+) because of the interaction between the electropositive B atoms and the anions of lithium salts. However, this effect can be largely impeded by the interaction between B and the electronegative ether oxygen atoms in the attached PEG molecular chains, which is called shielding effects. Enlarging the distance between the center B atoms and the electronegative ether oxygens is believed an effective way to reduce such shielding effect. In this work, different number of siloxane spacing groups (Si_x) are introduced in between the B atom and ether oxygen in the B-PEG macromolecules (B- Si_x -PEGs $x=0,1,3,5$). As the siloxane oxygen has less electronegativity to ether oxygen, the shielding effects should be weakened and the ability of B atoms to trap the anions of lithium salt will increase. As the introduction of siloxane groups into the B- Si_x -PEGs polymer electrolytes will also reduce the neat contents of both B and PEG, which has the opposite effect on the whole ionic conductivities and t_{Li}^+ values, both the ionic conductivities and t_{Li}^+ values of B- Si_x -

* Corresponding author D. Shi (deanshi2012@yahoo.com, deanshi2012@hubu.edu.cn), Z. Wei (weizy@hubu.edu.cn)

PEGs electrolytes first increase then decrease along with x increases from 0 to 5 when the $[\text{Li}^+]:[\text{EO}]$ mole ratio is fixed at 1:10. The peak value exists at $x=1$. Meanwhile, both the mobilities of Li^+ ions (confirmed via ^7Li nuclear magnetic resonance (NMR) results) and bis(trifluoromethane) sulfonimide (TFSI^-) anions (confirmed from Raman spectra analysis) get their maximum values when $x=1$. However, if both the concentrations of TFSI^- and B atoms in B- Si_x -PEGs electrolytes are taken into consideration, the normalized diffusion coefficient of TFSI^- anions obtained by molecular dynamic (MD) simulation decrease monotonically when x increases from 0 to 5. As the diffusion coefficient of TFSI^- anions is determined by the interaction probability and strength of TFSI^- and B atoms, these results confirm that the interaction between the B atoms and TFSI^- anions increase with x . Namely, the deshielding effects of siloxane spacing groups on B atoms increase along with the increase of x in the B- Si_x -PEGs polymer electrolytes, while it levels off when x exceeds 3.

Key Words: Shielding effect, B, siloxane groups, PEG, solid polymer electrolytes

Introduction

Compared with the safety problems caused by the flammability and toxicity of liquid electrolytes, Poly(ethylene glycol) (PEG) based solid polymer electrolytes have become potential candidates for next-generation battery on the basis of the merits including excellent safety, electrochemical stability and excellent electrochemical compatibility with lithium metal anode.^[1-4] However, its intrinsic low ionic conductivity, undesired lithium transference number (t_{Li}^+), poor wettability and low stability between electrodes have largely limited its wide application. Tremendous efforts have been made to conquer those puzzles and have already achieved series of achievements.^[5-10]

Recently, t_{Li}^+ of polymer electrolytes receives significant attention as it is the key point to solve the critical challenge about fast charging technology.^[11-14] Anion acceptors, which can interact with anions of lithium salts and suppress their movement abilities, are promising candidates to get high t_{Li}^+ .^[15] Boron (B) is one of the most widely used and studied anion acceptors. Many researches show that the incorporation of B atoms could trap anion^[16] and enhance the dissociation of Li salt and improve Li^+ transport property.^[17-24] However, some previous studies about B-PEG polymer electrolytes do not show much enhancement of t_{Li}^+ values.^[25-28] which indicates that the interactions between B atoms and the anions of lithium salt are shielded to some extent. And the interaction between the ether oxygen atoms in the attached PEG molecular chains and B atoms are believed the most possible reason for this shielding effect.^[28] However, there is no direct evidence to support this viewpoint so far.

In order to clearly address this question, different number of siloxane spacing groups with oxygen atoms having relatively weaker electronegativity^[29] are introduced in between the B atom and PEG molecule chains in the B-PEG polymer electrolytes in this work (B-Si_x-PEGs $x=0,1,3,5$). The macroscopic properties of the B-Si_x-PEGs polymer electrolytes, such as ionic conductivities and t_{Li}^+ values are characterized. Raman spectroscopy and ⁷Li nuclear magnetic resonance (NMR) spectroscopy have been carried out to investigate the ions mobility of lithium salt in the B-Si_x-PEGs electrolytes.

Meanwhile, molecular dynamics (MD) simulation^[30-33] is also used to calculate both the microscopic interaction force between B atom and the anion of bis(trifluoromethane) sulfonimide lithium salt (LiTFSI) and the macroscopic properties of the B-Si_x-PEGs polymer electrolytes, such as ionic conductivity and t_{Li}^+ value. From the macroscopic

point of view, the theoretical predictions fit very well with the experimental results. If both the concentrations of TFSI⁻ and B atoms in B-Si_x-PEGs electrolytes are taken into consideration, the normalized diffusion coefficient of TFSI⁻ anions obtained by molecular dynamic (MD) simulation confirms the deshielding effects of siloxane spacing groups on B atoms increase along with x in the B-Si_x-PEGs polymer electrolytes. The result of this work will help to give a clear guidance to prepare high performing PEG based polymer electrolytes for lithium ion battery.

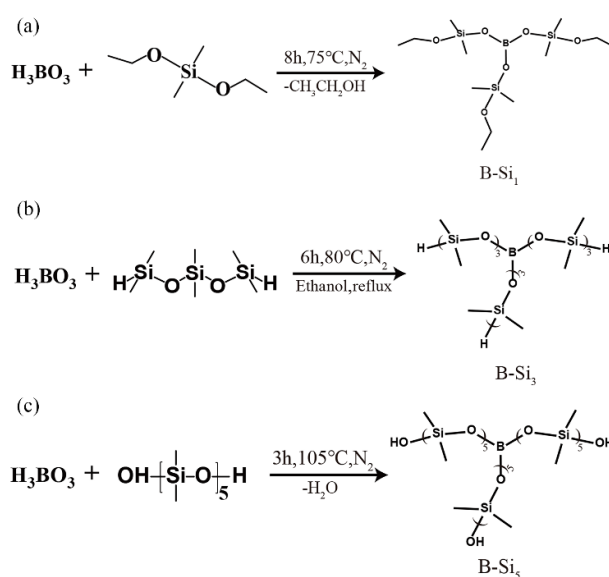
2.Experimental

2.1 Materials

PEG (Mn=400 g/mol) was purchased from Sino-pharm Chemical Reagent Co. Ltd., and MPEG (Mn=1000 g/mol) from Tokyo Chemical Industry Co. Ltd. Both were dried in vacuum at 110°C for 24h before use. Hexamethylene-1,6-diisocyanate homopolymer (HDI trimer) (27.2% in butyl acetate solvent) was received from Bayer Material Science. Hydroxyl terminated polydimethylsiloxane (Si₅) from Hubei Blue Sky New Materials INC, Tetrabutylammonium hydroxide, n-propyltrimethoxysilane (PTMS), boric acid (H₃BO₃), dimethyl diethoxysilane (Si₁), 1,1,3,3,5,5-Hexamethyltrisiloxane (Si₃), LiTFSI, Tris(pentafluorophenyl) borane, were obtained from Sigma-Aldrich. N,N-Dimethyl Formamide (DMF, 99.8%) was purchased from Aladdin. Olivine type LiFePO₄ (The average size of LiFePO₄ was about 800 nm) and acetylene black were purchased from Shenzhen Kejing Star Technology Co. Ltd. Polyvinylidene fluoride (PVDF Mw=1,000,000 g/mol) was received from Arkema. These materials were used as received. Tetrahydrofuran (THF) was refluxed with hydride (CaH₂) overnight, followed by reduced pressure distillation.

2.2 Synthesis of B-Si_x-PEGs oligomers

The B-Si_x-PEGs ($x=1,3,5$) was synthesized from the reaction of H₃BO₃ and three types of siloxanes (**B-Si_x for Scheme 1 and B-Si_x-PEGs for Scheme S1**). For B-Si₁-PEG, H₃BO₃ (0.618g, 0.01mol) and dimethyl diethoxysilane (4.448g, 0.03mol) were mixed in around-bottom flask equipped with stirring. The mixture was stirred vigorously at 75°C for 8h under N₂ atmosphere to eliminate the generated alcohol. The product was denoted as B-Si₁. Then PEG400 (12g, 0.03mol) was added under 100°C for 12h. For B-Si₃-PEG, using the dehydrocoupling method proposed by Ohetal and Schiavon et al. The H₃BO₃ (0.618g, 0.01mol), 1,1,3,3,5,5-Hexamethyl trisiloxane (6.254g, 0.03mol), Tris(pentafluorophenyl) borane (0.004g) were mixed in around-bottom flask equipped with stirring. The mixture was stirred vigorously at 80°C for 6h under N₂ atmosphere and the product was denoted as B-Si₃. Then PEG400 (12g, 0.03mol) was added under 100°C for 8h. For B-Si₅-PEG, H₃BO₃ (0.618g, 0.01mol) and hydroxyl terminated polydimethylsiloxane (11.64g, 0.03mol) were mixed in around-bottom flask equipped with stirring. The mixture was stirred vigorously at 105°C for 3h under N₂ atmosphere. The product was denoted as B-Si₅, Then PEG400 (12g, 0.03mol) was added under 100°C for 8h.



Scheme 1. Reaction scheme for the synthesis of B-Si_x

2.3 Preparation of B-Si_x-PEGs polymer electrolytes

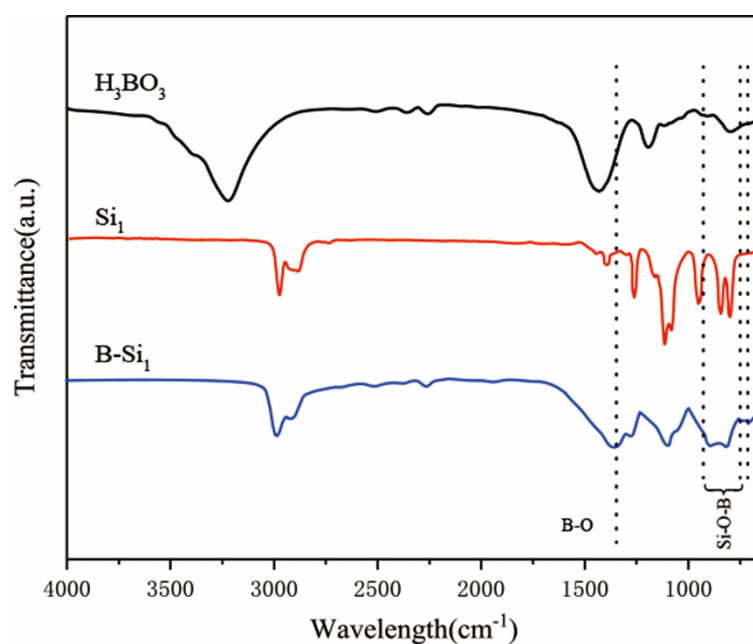
The preparation process of B-Si_x-PEGs polymer electrolytes was similar to our previous work,^[34] B-Si_x-PEGs (0.01mol), MPEG (0.01mol, 1g) and PTMS (4.92g, 0.02mol) were stirred in a 250 mL three neck flask for 2h at 80°C. After that, tetra-butyl ammonium hydroxide was added as catalyst to adjust the pH value to 9-10 and further stirred at 80°C. After 4h, the residual by-product methanol and catalyst were removed by reduced pressure distillation. Then HDI trimer and LiTFSI were added and further reacted for 1h. The ratio between PTMS and HDI trimer was 10:2 and [Li⁺]:[EO] mole ratio was controlled at 1:10. Finally, the homogeneous mixtures were casted on the Teflon substrate under 80°C, the ~80μm thick polymer electrolyte (B-Si_x-PEGs) membranes were obtained. The pictures of B-Si_x-PEGs electrolytes membranes are shown in **Figure S1**.

Characterizations

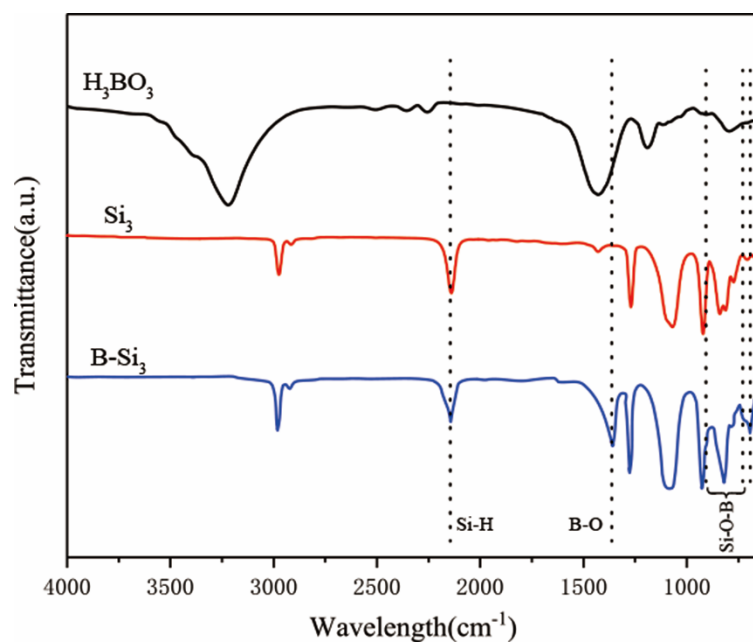
¹H NMR spectroscopy were carried out on a UNITY INOVA 600 MHz spectrometer (Varian Company Ltd.) at room temperature. The solid state ⁷Li NMR experiments were performed on Bruker advance iii 500MHz solid state NMR spectrometer operating at a ⁷Li resonance frequency of about 195MHz. The polymer electrolytes (approximately 500mg) were packed into sealed 5mm Pyrex tubes. All the samples were recorded with a RF pulse length of 2μs and a recycle delay of 60s. Fourier transform infrared (FT-IR) spectra measurements were recorded in the absorption mode on Thermo Fisher Scientific Nicolet iS50, the vibrational frequency ranges from 400 to 4000cm⁻¹ with resolution of 4cm⁻¹. Raman spectra were measured by the LabRAM HR Evolution Raman microscope system. All the spectra were recorded in the range between 730 and 760 cm⁻¹. The ionic conductivity of B-Si_x-PEGs was measured by electrochemical impedance spectroscopy (EIS) in the frequency range of 0.1-10⁶ Hz with 10 mV of AC amplitude via

electrochemical workstation (Shanghai Chenhua Instrument Co., Ltd). Samples were prepared for measurement by sandwiching B-Si_x-PEGs between two stainless-steel electrodes. The ionic conductivity of the electrolyte could be calculated from the following equation: $\sigma = (1/R) \times (d/A)$. Where R is the bulk resistance, L is Electrolyte membrane thickness, A represents effective electrode area. Li⁺ ion transference number (t_{Li}^+) was carried out by using DC polarization and AC impedance combination method. The B-Si_x-PEGs were sandwiched between two regular lithium metal to form asymmetrical Li/B-Si_x-PEGs/Li cell. The cell was polarized by a constant DC voltage of 10 mV(ΔV). t_{Li}^+ values were determined by following equation: $t_{Li}^+ = \frac{I_{SS}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{SS} R_{SS})}$. Where I_0 and I_{SS} are the initial current and the steady-state current, R_0 and R_{SS} are the initial and steady-state interfacial resistances.

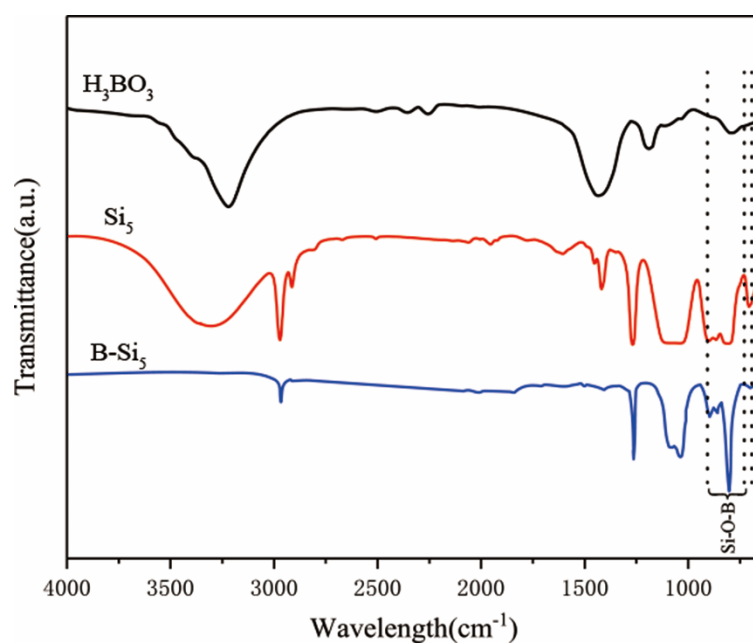
3. Results and Discussion



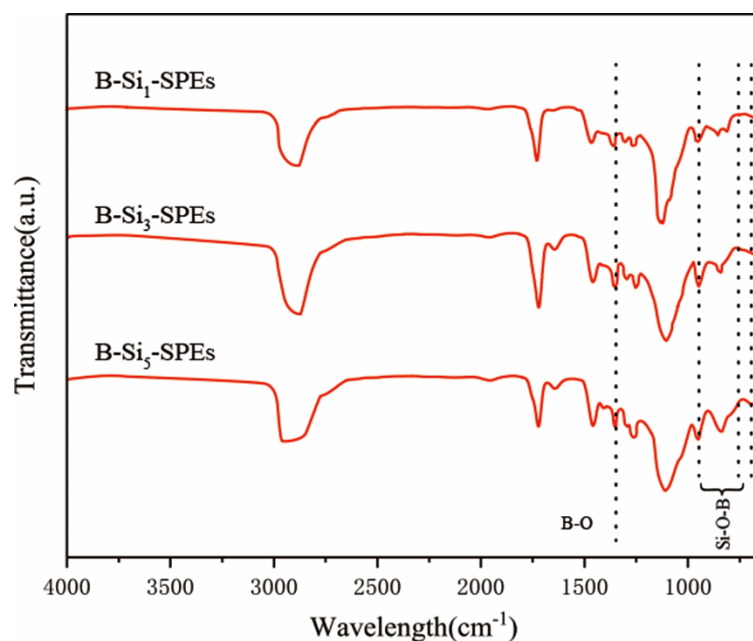
(a)



(b)



(c)



(d)

Figure 1. FTIR spectra of (a) H_3BO_3 , Si_1 , B-Si_1 (b) H_3BO_3 , Si_3 , B-Si_3 , (c) H_3BO_3 , Si_5 , B-Si_5 and (d) B-Si_x -PEGs ($x=1,3,5$).

FTIR spectra of H_3BO_3 , Si_x , B-Si_x and B-Si_x -PEGs are illustrated in **Figure 1**. Comparing the spectra of H_3BO_3 and B-Si_x , the disappearance of the peaks at around 3211cm^{-1} in H_3BO_3 indicates the complete reaction of the hydroxyl group in H_3BO_3 . Peak at 1330cm^{-1} is associated with B-O stretching vibrations. The new present peaks 695cm^{-1} , 673cm^{-1} and 883cm^{-1} in B-Si_x series confirm the formation of B-O-Si bonds (**Figure 1a-c**).^[35, 36] In **Figure 1d**, all the B-Si_x -PEGs samples have similar spectra and there are no obvious -OH peaks above 3000cm^{-1} , which indicate the complete reaction between siloxane group in PTMS and -OH groups from PEG and MPEG.

$^1\text{H-NMR}$ is further used to confirm the formation of B-Si_x (**Figure 2** and **Figure S2**). **Figure 2** is the $^1\text{H-NMR}$ spectrum of B-Si_1 . The peaks **a** and **c** can be assigned to the protons of the $-\text{Si-CH}_3-$ and $-\text{CH}_2-\text{CH}_3-$, respectively. And peak **b** is attributed to the protons of $-\text{CH}_2-$. This result fit very well with the B-Si_1 molecular structure. Similar results can be found in **Figure S2** for B-Si_3 and B-Si_5 .

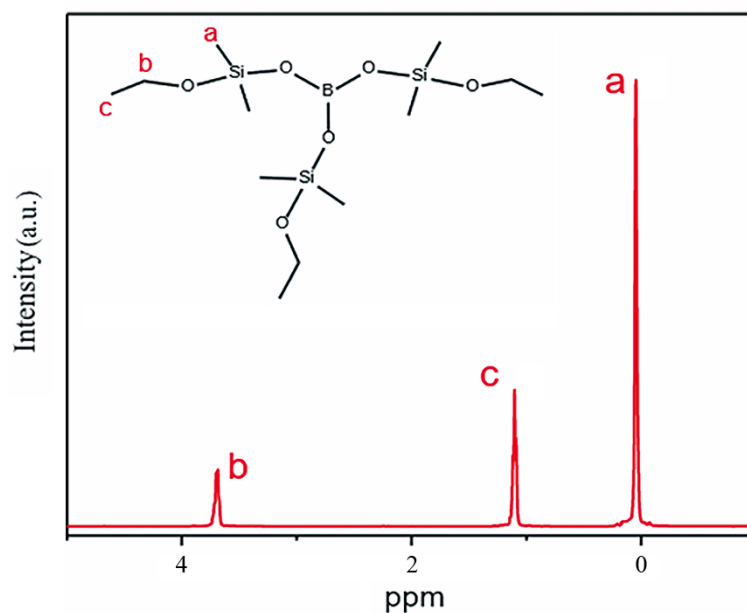
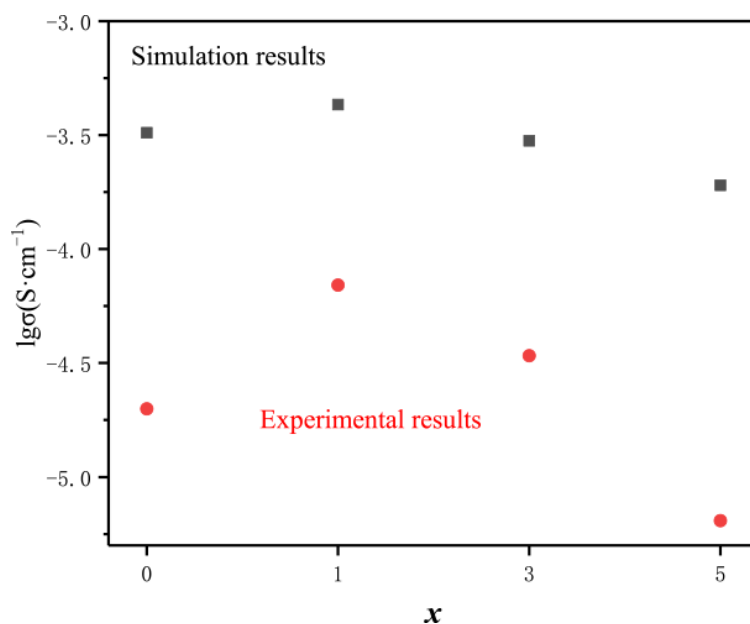
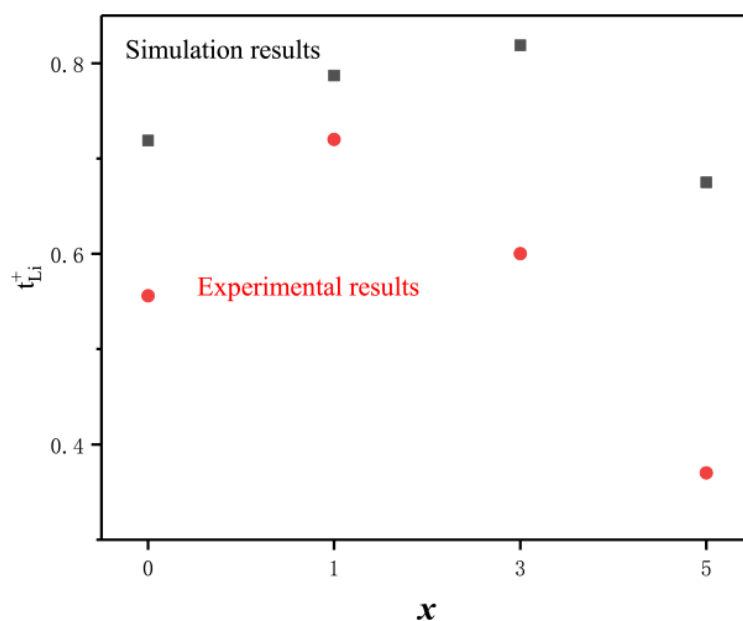


Figure 2. ^1H NMR spectra of B-Si₁

Figure 3 illustrates the ionic conductivities and t_{Li}^+ values of different polymer electrolytes at 30°C. As shown in **Figure 3**, along with the increasing of siloxane spacing groups in the B-Si_x-PEGs electrolytes, both the experimental ionic conductivities and t_{Li}^+ values get a peak value at $x=1$. Cycle performance and C-rates of the batteries assembled with Li/B-Si_x-PEGs/LiFePO₄ also get its best properties when B-Si₁-PEGs is applied as the electrolyte. (See **Figure S3** and **Figure S4** in Supporting Information).



(a)



(b)

Figure 3 Experimental and simulation results of B-Si_x-PEGs electrolytes at 30°C (a) ionic conductivities and (b) t_{Li^+} values

The ^7Li NMR and Raman spectra of B-Si_x-PEGs shown in **Figure 4** and **Figure 5** give further evidence to support the above results. ^7Li NMR is usually used to investigate the interaction between the electrolyte and Li^+ .^[37-40] If the chemical shift of the free Li^+ ion in D_2O solution is assigned to be 0 ppm, all the chemical shifts of Li^+ ions that are less free than Li^+ ion in D_2O will move to upfield. The lower the mobility of the ions is, the higher upfield of the chemical shifts will be. As shown in **Figure 4**, the chemical shifts of ^7Li of B-Si_x-PEGs first decrease when x varies from 0 to 1, and then increase with increasing x from 1 to 5 in B-Si_x-PEGs electrolytes. This result implies that the freedom of Li^+ ions in B-Si₁-PEGs is the highest. On the other hand, the full width at half maximum (the width of a spectrum curve measured between those points on the y-axis which are half the maximum amplitude, ***FWHM***) is also well-known as a reasonable rule to reflect the ion mobility in solid-state electrolyte system.^[41, 42] The smaller the ***FWHM*** is, the higher the mobility of Li^+ ion will be. As shown in **Table 1**, the ***FWHM*** of B-Si₁-PEGs is also the smallest

one.

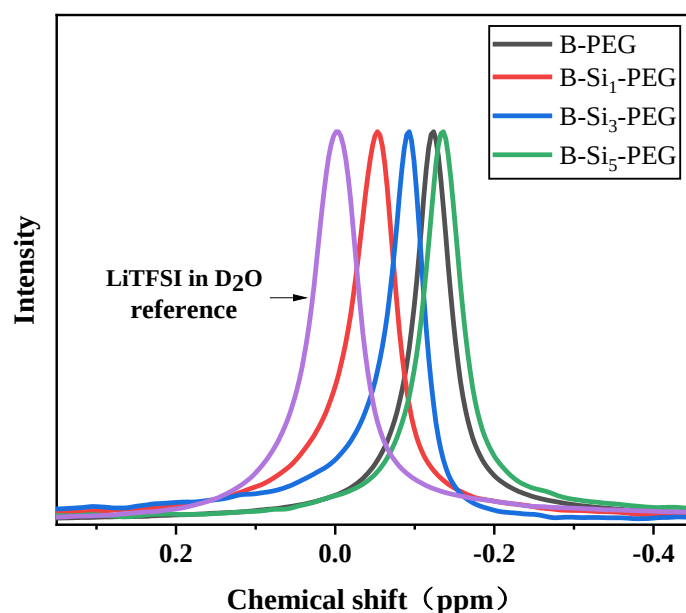


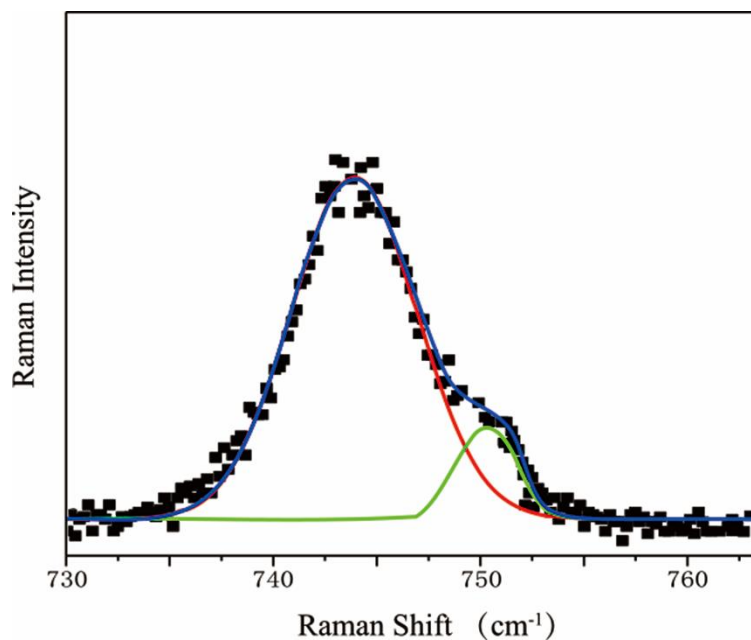
Figure 4. ^7Li NMR spectra of B-Si_x-PEGs electrolytes. (δ referenced to LiTFSI in D₂O)

Table 1 The *FWHM* of ^7Li peak in B-Si_x-PEG s electrolytes

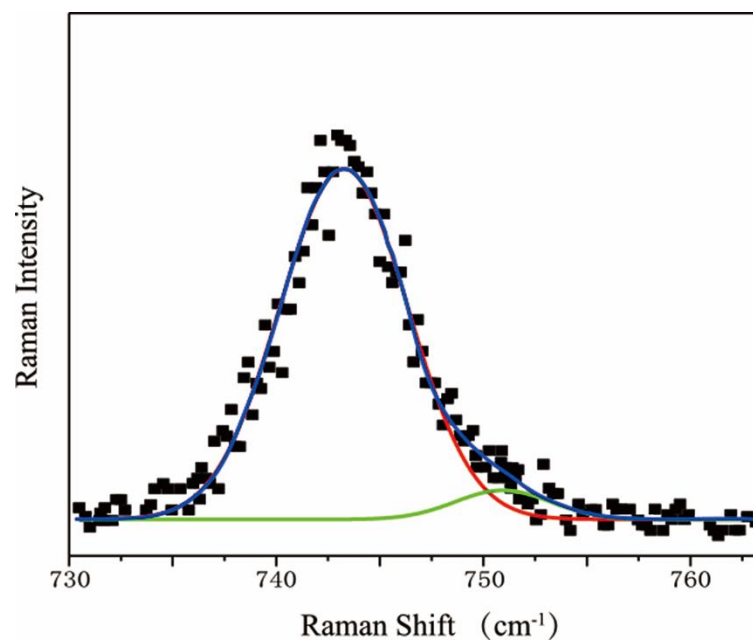
Sample	B-Si ₀ -PEG	B-Si ₁ -PEGs	B-Si ₃ -PEGs	B-Si ₅ -PEGs
FWHM(Hz)	10.82	9.58	10.21	12.15

Meanwhile, Raman spectra of LiTFSI/B-Si_x-PEGs systems shown in **Figure 5** can give some hints of the mobility of TFSI⁻ anions. It is reported that in LiTFSI/PEO systems, the Raman signal of δ_{CF_3} band around 740~750cm⁻¹ will be split into two peaks. The peak near 740cm⁻¹ is attributed to the dissociated or free TFSI⁻ anions, while the one near 750cm⁻¹ represents those coordinated or trapped TFSI⁻ anions.^[43-46] **Figure 5** displays the Raman spectra from 730~760cm⁻¹ where 742cm⁻¹ (red line) should be ascribed to the free TFSI⁻ anions and 748cm⁻¹ (green line) represents those coordinated or trapped ones TFSI⁻. It is believed that the portion of free TFSI⁻ anions (φ) should be proportional to its mobility, where φ can be calculated via the peak area of 742cm⁻¹ and 748cm⁻¹ ($\varphi =$

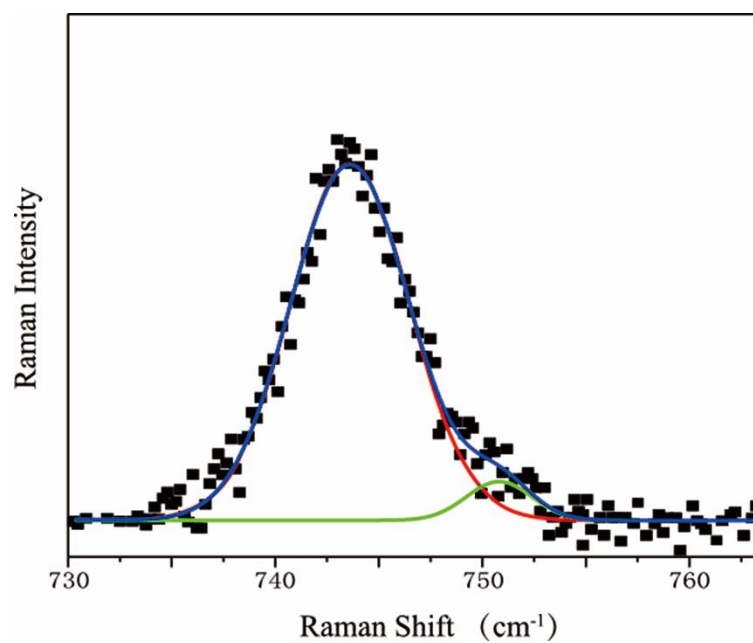
$\frac{A_{742}}{A_{748}+A_{742}}$). As shown in **Figure 5e**, the mobility of TFSI⁻¹ anions also reaches its maximum value at $x=1$.



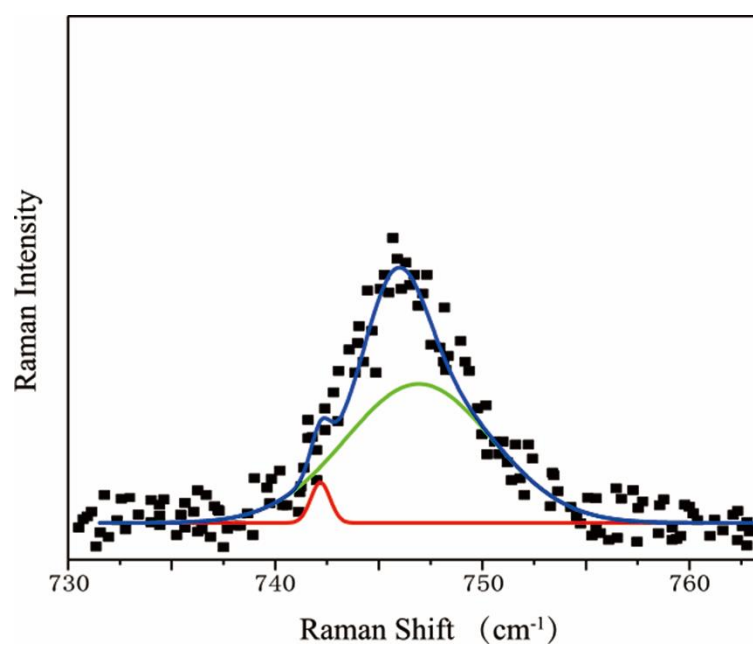
(a)



(b)



(c)



(d)

Accepted Manuscript

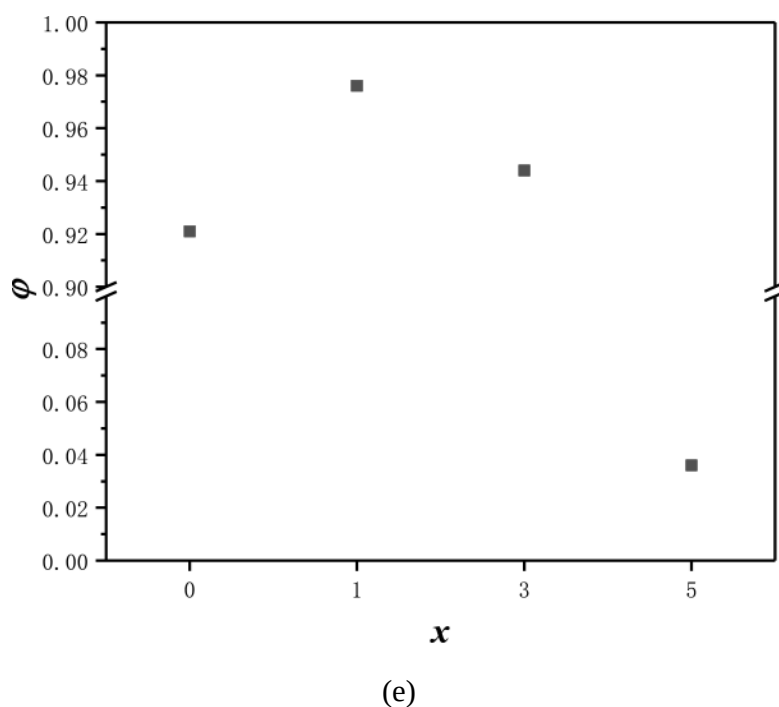


Figure 5. Raman spectra of (a-d) LiTFSI/B-Si_x-PEGs series; (e) The portion of free TFSI⁻¹ anions

However, all these results seem unable to be explained by the enhancement of the deshielding effect of B atoms along with the increase of x number in B-Si_x-PEGs. If the shielding effects of ether oxygen in PEG on the center B atoms are weakened, the interaction between B and TFSI⁻ anions should be enhanced, which will decrease the mobility of TFSI⁻ anions. On the other hand, the mobility of Li ions should be increased. In this case, at least the values of t_{Li}^+ should increase monotonically along with the increase of x value, rather than get its maximum at $x=1$. It is well known that in B-Si_x-PEGs electrolytes, Li⁺ cations are coordinated with ether oxygen in PEG macromolecular chains and TFSI⁻ anions interact with B atoms. The dissociation degree of LiTFSI salt as well as the mobility of Li⁺ and TFSI⁻ ions are determined by these two interactions simultaneously. Therefore, before we can study the effects of siloxane group numbers on the interaction strength between B and TFSI⁻, both the dissociation degree of LiTFSI salt and the interaction between Li⁺ and PEG should be clearly characterized.

In this case, molecular dynamic (MD) simulation (detailed methodology is illustrated in the

Supporting Information) method is applied. In the simulated systems, the LiTFSI salt is fully dissociated. The numbers of B atoms, Li^+ and TFSI $^-$ ions are fixed as 10, 30 and 30 in each system, respectively (**Table S1**). It is interesting that the simulated ionic conductivities and t_{Li}^+ (**Figure 3 and Table S2**) also give the same varying trends as the experimental ones, although their absolute values are different. Furthermore, the simulated diffusion coefficients for both Li^+ (D_{Li}^+) and TFSI $^-$ (D_{TFSI}^-) ions (**Table 2**) also get their highest value at $x=1$, which is in accordance with the experimental results. However, according to our previous assumption, D_{Li}^+ as well as t_{Li}^+ value should increase and D_{TFSI}^- should decrease monotonically along with the increase of x value. Here, it should be noted that LiTFSI salt in all the B-Si $_x$ -PEGs electrolyte systems considered in the simulation procedures are 100% dissociated. However, this is not the truth in real LiTFSI/B-Si $_x$ -PEGs system. When the $[\text{Li}^+]:[\text{EO}]$ mole ratio is fixed, the dissociation degree of LiTFSI salt might decrease along with the increase of siloxane groups. This is why the simulated peak value for t_{Li}^+ is at $x=3$ (**Figure 3b**), rather than $x=1$ for experimental result. Furthermore, the simulated value of D_{TFSI}^- at $x=5$ is even higher than that at $x=3$ (**Table 2**).

Table 2 Diffusion coefficient of Li^+ and TFSI $^-$ in B-Si $_x$ -PEGs electrolytes (m^2/s)

System	B-Si $_0$ -PEG	B-Si $_1$ -PEG	B-Si $_3$ -PEG	B-Si $_5$ -PEG
$D_{\text{Li}}^+ \times 10^{12}$	3.383	5.483	4.833	2.983
$D_{\text{TFSI}}^- \times 10^{12}$	1.323	1.483	1.067	1.433

In order to clearly understand all these deviations, the concentration variations of B and PEG as well as LiTFSI in different B-Si $_x$ -PEGs electrolytes should be taken into consideration, because the neat contents of both PEG and B will decrease along with the increase of added siloxane groups. While, the coordination of Li^+ cations with PEG ether oxygen and B/TFSI $^-$ interaction finally determine the mobility of Li^+ and TFSI $^-$ ions in B-Si $_x$ -PEGs electrolytes. Before the effects of these two kinds of interactions in different B-Si $_x$ -PEGs electrolytes can be investigated, the diffusion coefficients of Li^+ and TFSI $^-$ ions

should be normalized according to the neat contents of B, PEG and lithium salt. And all other parameters, such as ionic conductivities and t_{Li}^+ values can be calculated from the according normalized diffusion coefficients. In order to avoid the effect of the dissociation degree of LiTFSI salt in different systems, values obtained by simulation are used.

It is well known that D_{Li}^+ in the electrolyte solution is determined by both the concentration of Li^+ ions and electrolyte. Higher the Li^+ concentration^[47-49] or lower electrolyte concentration will all result in lower the D_{Li}^+ .^[50] Along with the increasing the siloxane groups in B-Si_x-PEGs electrolytes, both the concentrations of Li^+ ions and PEG decrease, which will give opposite effect of D_{Li}^+ . Although this mutual counteraction cannot be fully cancelled out, it is reasonable to assume that the concentration variation of Li^+ ions and PEG has no effect on the value of D_{Li}^+ . In other words, there is no need for normalization for D_{Li}^+ in different B-Si_x-PEGs electrolytes. However, the concentration decreasing of TFSI⁻ and B atoms are all benefit to the increase of D_{TFSI}^- in B-Si_x-PEGs electrolytes. Therefore, normalization of the simulated D_{TFSI}^- values should be done before we can use them to predict the interaction strength between B and TFSI⁻ anions. If we use $x=0$ as a reference, the concentrations of B and TFSI⁻ in B-Si₀-PEGs are denoted as C_{0B} and C_{0TFSI}^- . The normalized diffusion coefficient (D_{TFSI}^{n-}) can be calculated by **Equation (1)**.

$$D_{TFSI}^{n-} = D_{TFSI}^- \left(\frac{C_{xB}}{C_{0B}} \right) \left(\frac{C_{xTFSI}^-}{C_{0TFSI}^-} \right) = D_{TFSI}^- \left(\frac{C_{xB}}{C_{0B}} \right)^2 \quad (1)$$

$$t_{Li}^{n+} = \frac{D_{Li}^+}{D_{TFSI}^{n-} + D_{Li}^+} \quad (2)$$

The normalized diffusion coefficient (D_{TFSI}^{n-}) and related t_{Li}^{n+} (calculated via **Equation (2)**) are shown in **Figure 6**. It can be found that D_{TFSI}^{n-} values decrease monotonically along with the increase of x value in B-Si_x-PEGs electrolytes, which indicates that the insertion of siloxane spacing groups between B and PEG actually alleviates the shielding effects of

ether oxygen atoms in PEG molecules on B atoms and enhances the interaction between B atoms and TFSI⁻ anions. From the t_{Li}^{n+} curve in **Figure 6b**, it can be found that it also increases monotonically along with x , while the deshielding effect levels off when x exceeds 3.

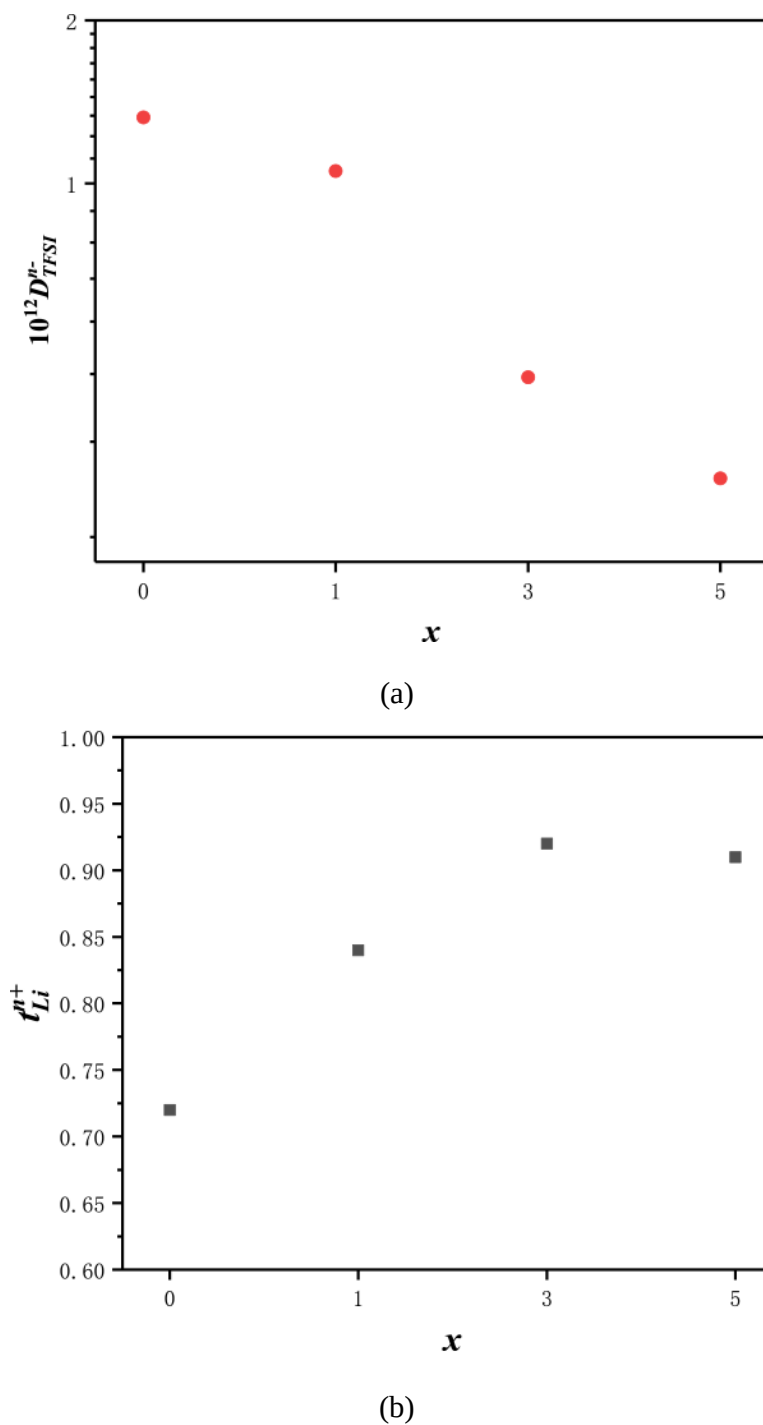


Figure 6 Variation of (a) normalized diffusion coefficient (D_{TFSI}^{n-}) and (b) t_{Li}^{n+} with x in B-Si_x-PEGs electrolytes

Figure 7 and **Figure S7** schematically illustrates shielding and deshielding effect of jointed atoms on B in B-Si_x-PEGs electrolytes. The shielding effect of the ether oxygens in PEG chains push away the TFSI⁻ anions from B atom in B-PEGs. While the oxygen in siloxane group largely alleviate the shielding effect and induce relatively stronger interaction and lower distance between B and TFSI⁻ anion in B-Si₁-PEGs.

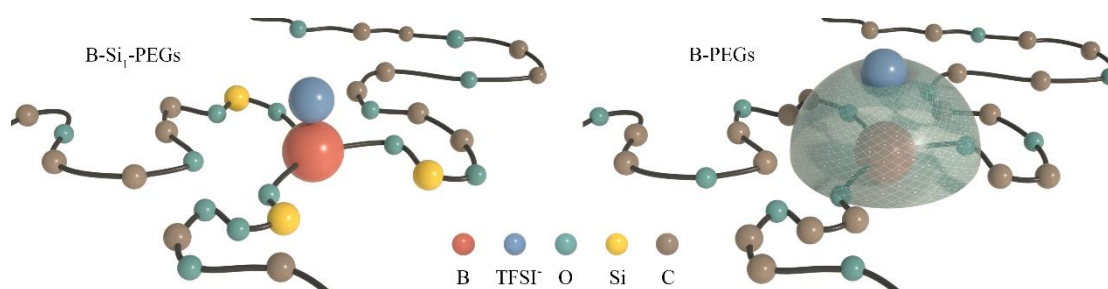


Figure 7. Illustration of the shielding and deshielding effects in LiTFSI/B-Si_x-PEGs electrolyte ($x=0$ and 1).

4. Conclusions

In this study, the shielding effect of the neighboring oxygen atoms on B in series of B-Si_x-PEGs polymer electrolytes are investigated. Compared to ether oxygen atoms, the siloxane oxygen atoms with less electronegativity have relatively weaker interaction with B atoms, which reduces the shielding effect on B atoms. If the concentrations differences of B and lithium salt in different B-Si_x-PEGs polymer electrolytes are taken into consideration, the deshielding effect increases along with the increase of the number siloxane oxygen atoms. And it levels off when the siloxane number exceeds 3. However, in the real electrolyte systems, as the increase of siloxane groups will reduce the neat content of both PEG and B in the B-Si_x-PEGs polymer electrolytes, which has opposite effect on ionic conductivity and t_{Li}^+ , the best macroscopic electrochemical properties exists at $x=1$. The Raman and ⁷Li-NMR results as well as the cycle

performance and capacity retention at different C-rate of LiFePO₄/B-Si_x-PEGs/Li half cells give the same conclusion. This work confirms the existence of shielding effect of ether oxygen atoms on the doped Lewis acidic B atoms in B-PEG and proposes a possible way to prepare high performance polymer electrolytes.

Acknowledgement

This work was supported by the NSFC No. 51973052, 51473047, 51903075, National Key Research and Development Program of China (2020YFC1106900) and Hubei Key Project of Science and Technology (CXZD2017000293). We also express our sincere thanks to the Chemcloudcomputing of Beijing University of Chemical Technology for the simulation work.

References

- [1] K. Xu, *Chem. Rev.* **2004**, *104*, 4303-4418.
- [2] L. Long, S. Wang, M. Xiao, Y. Meng, *J. Mater. Chem. A* **2016**, *4*, 10038-10069.
- [3] M. Winter, B. Barnett, K. Xu, *Chem Rev.* **2018**, *118*, 11433-11456.
- [4] J. Lopez, D. G. Mackanic, Y. Cui, Z. Bao, *Nat. Rev. Mater.* **2019**, *4*, 312-330.
- [5] Z. Xue, D. He, X. Xie, *J. Mater. Chem. A* **2015**, *3*, 19218-19253.
- [6] Q. Zhang, K. Liu, F. Ding, X. Liu, *Nano Res.* **2017**, *10*, 4139-4174.
- [7] A. Manthiram, X. Yu, S. Wang, *Nat. Rev. Mater.* **2017**, *2*.
- [8] Y. Wang, W.-H. Zhong, *ChemElectroChem* **2015**, *2*, 22-36.
- [9] L. Fan, S. Wei, S. Li, Q. Li, Y. Lu, *Adv. Energy Mater.* **2018**, *8*.
- [10] Y. He, C. Lu, S. Liu, W. Zheng, J. Luo, *Adv. Energy Mater.* **2019**.
- [11] Y. Liu, Y. Zhu, Y. Cui, *Nat. Energy* **2019**, *4*, 540-550.
- [12] H. Zhang, F. Chen, O. Lakuntza, U. Oteo, L. Qiao, M. Martinez-Ibañez, H. Zhu, J. Carrasco, M. Forsyth, M. Armand, *Angew. Chem., Int. Ed.* **2019**, *58*, 12070-12075.
- [13] G. L. Zhu, C. Z. Zhao, J. Q. Huang, C. He, J. Zhang, S. Chen, L. Xu, H. Yuan, Q. Zhang, *Small* **2019**, *15*, e1805389.
- [14] L. Chen, W. Li, L. Z. Fan, C. W. Nan, Q. Zhang, *Adv. Funct. Mater.* **2019**, *29*.
- [15] H. Zhang, C. Li, M. Piszcz, E. Coya, T. Rojo, L. M. Rodriguez-Martinez, M. Armand, Z. Zhou, *Chem. Soc. Rev.* **2017**, *46*, 797-815.
- [16] F. Jäkle, *Coord. Chem. Rev.* **2006**, *250*, 1107-1121.
- [17] J. Xu, J. Li, Y. Zhu, K. Zhu, Y. Liu, J. Liu, *RSC Adv.* **2016**, *6*, 20343-20348.

- [18] R. Tan, R. Gao, Y. Zhao, M. Zhang, J. Xu, J. Yang, F. Pan, *ACS Appl. Mater. Interfaces* **2016**, *8*, 31273-31280.
- [19] L. Yang, Z. Wang, Y. Feng, R. Tan, Y. Zuo, R. Gao, Y. Zhao, L. Han, Z. Wang, F. Pan, *Adv. Energy Mater.* **2017**, *7*.
- [20] S. Chen, F. Feng, Y. Yin, X. Lizo, Z. Ma, *Energy Storage Materials*. **2018**.
- [21] P. Yuan, C. Cai, J. Tang, Y. Qin, M. Jin, Y. Fu, Z. Li, X. Ma, *Adv. Funct. Mater.* **2016**, *26*, 5930-5939.
- [22] Y. Zhu, S. Xiao, Y. Shi, Y. Yang, Y. Hou, Y. Wu, *Adv. Energy Mater.* **2014**, *4*.
- [23] J. L. Schaefer, D. A. Yanga, L. A. Archer, *Chem. Mater.* **2013**, *25*, 834-839.
- [24] B. M. Savoie, M. A. Webb, T. F. Miller, III, *J. Phys. Chem. Lett.* **2017**, *8*, 641-646.
- [25] N.-S. Choi, S.-W. Ryu, J.-K. Park, *Electrochim. Acta* **2008**, *53*, 6575-6579.
- [26] N.-S. Choi, Y. M. Lee, K. Y. Cho, D.-H. Ko, J.-K. Park, *Electrochem. Commun.* **2004**, *6*, 1238-1242.
- [27] J. Shim, J. S. Lee, J. H. Lee, H. J. Kim, J. C. Lee, *ACS Appl. Mater. Interfaces* **2016**, *8*, 27740-27752.
- [28] D. Li, X. Ji, X. Gong, F. Tsai, Q. Zhang, L. Yao, T. Jiang, R. K. Y. Li, H. Shi, S. Luan, D. Shi, *J. Power Sources* **2019**, *423*, 349-357.
- [29] F. Weinhold, R. West, *Organometallics* **2011**, *30*, 5815-5824.
- [30] O. Borodin, G. D. Smith, *Macromolecules* **2006**, *39*, 1620-1629.
- [31] C. Park, M. Kanduć, R. Chudoba, A. Ronneburg, S. Risse, M. Ballauff, J. Dzubiella, *J. Power Sources* **2018**, *373*, 70-78.
- [32] Y. Feng, R. Tan, Y. Zhao, R. Gao, L. Yang, J. Yang, H. Li, G. Zhou, H. Chen, F. Pan, *Nanoscale* **2018**, *10*, 5975-5984.
- [33] H. Wu, C. D. Wick, *Macromolecules* **2010**, *43*, 3502-3510.
- [34] X. Ji, H. Zeng, X. Gong, F. Tsai, T. Jiang, R. K. Y. Li, H. Shi, S. Luan, D. Shi, *J. Mater. Chem. A*. **2017**, *5*, 24444-24452.
- [35] X. Zhao, C. Zang, Y. Sun, K. Liu, Y. Wen, Q. Jiao, *J. Mater. Sci.* **2017**, *53*, 1167-1177.
- [36] K. J. Sreejith, P. V. Prabhakaran, K. P. Laly, R. Dimple, S. Packirisamy, *Ceram. Int.* **2016**, *42*, 15285-15293.
- [37] B. Qiao, G. M. Leverick, W. Zhao, A. H. Flood, J. A. Johnson, Y. Shao-Horn, *J. Am. Chem. Soc.* **2018**, *140*, 10932-10936.
- [38] H. A. Every, F. Zhou, M. Forsyth, D. R. MacFarlane, *Electrochim. Acta* **1998**, *43*, 1465-1469.
- [39] S. Wong, R. A. Vaia, E. P. Giannelis, D. B. Zax, *Solid State Ionics* **1996**, *86-88*, 547-557.
- [40] N. Wu, P. H. Chien, Y. Qian, Y. Li, H. Xu, N. S. Grundish, B. Xu, H. Jin, Y. Y. Hu, G. Yu, J. B. Goodenough, *Angew. Chem., Int. Ed. Engl.* **2020**, *59*, 4131-4137.
- [41] N. Lago, O. Garcia-Calvo, J. M. Lopez del Amo, T. Rojo, M. Armand, *ChemSusChem* **2015**, *8*, 3039-3043.
- [42] Y. Yan, R.-S. Kühnel, A. Remhof, L. Duchêne, E. C. Reyes, D. Rentsch, Z. Łodziana, C. Battaglia, *Adv. Energy Mater.* **2017**, *7*.
- [43] L. Edman, *J. Phys. Chem. B*. **2000**, *104*, 7254-7258.
- [44] D. Brouillette, D. E. Irish, N. J. Taylor, G. r. Perron, M. Odziemkowski, J. E. Desnoyers, *Phys. Chem. Chem. Phys.* **2002**, *4*, 6063-6071.
- [45] A. Bakker, J. Lindgren, K. Hermansson, *Polymer* **1996**, *37*, 1871-1878.
- [46] A. Ferry, *J. Electrochem. Soc.* **1998**, *145*.
- [47] L. Tang, *Cem. Concr. Res.* **1999**, *29*, 1463-1468.
- [48] A. Z. Akcasu, *Polymer* **1981**, *22*, 1169-1180.
- [49] V. Lesch, Z. Li, D. Bedrov, O. Borodin, A. Heuer, *Phys. Chem. Chem. Phys.* **2016**, *18*, 382-392.

[50] R. Bhandary, M. Schönhoff, *Electrochim. Acta* **2015**, *174*, 753-761.

For Table of Contents Use Only

The shielding effects of ether oxygen atoms in PEG molecular chains on the B atoms can be alleviated by introducing siloxane oxygens in between (B-Si_x-PEGs). the deshielding effects of siloxane spacing groups on B atoms increase along with the increase of *x* in the B-Si_x-PEGs polymer electrolytes.

